

CondorBrain Training Dashboard

Technical Operations Report

1. Introduction

The CondorBrain Training Dashboard provides a real-time operational view of CondorNet model training. This report presents a structured, formal description of each visualization, metric, and diagnostic component. The objective is to enable quantitative researchers, ML engineers, and system operators to interpret training behavior with precision and to identify early indicators of model health, instability, or convergence.

2. Fuzzy Gate Activation Heatmap

The Fuzzy Gate Heatmap visualizes the instantaneous activation levels of CondorNet’s predicate gates. These gates represent interpretable trading conditions derived from both domain-informed rules and learned patterns.

2.1 Activation Scale

Gate activations are continuous values in the interval $[0, 1]$, rendered using a perceptual color gradient:

Activation	Color	Interpretation
0.0	Dark (black/purple)	Gate inactive; condition not satisfied
0.5	Mid-tone (orange)	Near threshold; condition uncertain
1.0	Bright (yellow/white)	Gate strongly active; condition satisfied

The use of a **soft sigmoid** with steepness parameter $\alpha = 20$ ensures smooth transitions and early visibility of near-trigger states.

2.2 Predicate Definitions

The first five gates correspond to interpretable, domain-aligned rules:

Gate	Name	Learned Rule	Trading Interpretation
0	Volatility Spike	$IVR > 75$	High-volatility environment favorable for premium selling
1	Liquidity Lock	$\frac{\text{Spread}}{\text{Price}} > 1.12\%$	Wide spreads; block entries to avoid slippage

2	Trend Reversal	RSI < 25	Oversold condition; potential mean reversion		
3	Gap Guard	1-minute price jump > 1.12%	Sudden discontinuity; increase caution		
4	Gamma Hedge	\(	\Gamma	> 0.011 \)	High gamma exposure; hedge recommended
5–9	Learned Predicates	Data-driven	Model-discovered structural patterns		

2.3 Interpreting Temporal Patterns

- **Columns** represent sequential time steps (newest on the right).
- **Rows** correspond to individual gates.
- **Persistent bright bands** indicate sustained activation of a specific condition.
- **Vertical bright clusters** indicate multi-condition alignment—often high-signal trading moments.
- **Dark regions** reflect calm or low-signal market regimes.

3. Loss Metrics Overview

The dashboard tracks **12 distinct loss components**, each displayed with a live sparkline. These metrics collectively quantify prediction accuracy, trading performance, model stability, and regularization behavior.

3.1 Primary Loss

Metric	Description	Objective
loss	Weighted sum of all components	Minimize

3.2 Prediction Quality

Metric	Description	Objective
mse	Mean Squared Error between predictions and targets	Minimize

3.3 Trading Performance Metrics

Metric	Description	Objective
npdd	Normalized Profit per Dollar of Drawdown	Maximize
sharpe	Annualized Sharpe Ratio	Maximize
dd	Maximum drawdown	Minimize
turnover	Trading frequency penalty	Minimize
growth	Capital growth trajectory	Maximize

3.4 Regularization and Stability Metrics

Metric	Description	Objective
fuzzy	Fuzzy gate variance; encourages consistent gating	Maintain stability
pattern_ent	Pattern entropy; promotes pattern diversity	Lower is better

group_inv	Group invariance penalty	Minimize
rho	Spectral radius of dynamics matrix A	Maintain ≈ 0.01
energy	Hidden-state energy	Minimize

3.5 Sparkline Interpretation

- **Downward trends** in loss, mse, dd, turnover → desirable.
- **Upward trends** in sharpe, npdd, growth → desirable.
- **Flat lines** may indicate convergence or stagnation.
- **High-frequency oscillations** may indicate excessive learning rate.

4. Loss Trajectory Visualization

The Loss Trajectory Chart provides a streaming view of all loss components over training steps.

4.1 Chart Structure

- **X-axis:** Global training step.
- **Y-axis:** Loss magnitude.
- **Multiple colored lines:** Individual loss components.

4.2 Component Color Coding

Component	Color	Priority
loss	Blue	Primary
mse	Orange	Accuracy
sharpe	Green	Performance
npdd	Red	Risk-adjusted return
dd	Purple	Drawdown control

4.3 Healthy Training Patterns

- Rapid initial decline → effective early learning.
- Gradual descent → stable convergence.
- Plateaus → may require LR adjustment.
- Temporary spikes → normal recovery from noisy batches.

5. Diagnostics Panel

The Diagnostics Panel provides real-time insight into optimizer behavior and numerical stability.

5.1 Learning Rate

- **Decreasing curve:** Cosine annealing (default).
- **Flat:** Constant LR.
- **Stepwise drops:** Step-schedule.

5.2 Gradient Norm

Range	Interpretation
	Possible vanishing gradients
0.1–1.0	Healthy
1.0–10.0	Elevated; monitor
> 10.0	Clipping active

Gradient clipping at **1.0** prevents runaway updates.

5.3 AMP Scaler (Mixed Precision)

Value	Interpretation
65536	Initial scale
Decreasing	Overflow detected; scaler adjusting
Stable	Healthy
Very low	Potential numerical instability

6. Training Progress Header

The header summarizes global training progress.

6.1 Displayed Elements

- Epoch counter
- Global step
- Circular progress indicator
- Linear progress bar
- Estimated time remaining (ETA)

6.2 ETA Computation

$$\text{ETA} = \frac{\text{elapsed time}}{\text{steps completed}} \times \text{steps remaining}$$

Updated every 10 batches.

7. WebSocket Telemetry

The dashboard receives real-time updates via WebSocket.

7.1 Connection States

- **Connected** — receiving live data
- **Reconnecting** — transient interruption
- **Disconnected** — backend unavailable

7.2 Telemetry Frequency

Endpoint	Frequency	Data
/telemetry/step	Every 10 batches	Loss components
/telemetry/status	Every 10 batches	Progress, ETA
/telemetry/fuzzy	Every 10 batches	Gate activations
/telemetry/epoch	End of epoch	Summary
/telemetry/complete	End of training	Final results

8. Training Phase Interpretation

8.1 Epoch 1 — Initialization

- High or negative loss (Sharpe component).
- Heatmap appears random.
- Rapid early improvement expected.

8.2 Epochs 2–3 — Learning Phase

- Significant loss reduction.
- Sharpe improves sharply.
- Gate patterns begin to emerge.

8.3 Epochs 4–5 — Refinement

- Slower improvements.
- Heatmap stabilizes into recognizable patterns.
- Monitor for overfitting.

8.4 Indicators of Healthy Training

- Loss decreasing
- Sharpe increasing
- Drawdown stable and low
- Gate patterns non-random
- Gradient norm stable (0.1–1.0)

8.5 Indicators of Training Issues

- Loss oscillation or increase
- Gradient norm collapsing or clipping constantly
- Heatmap stuck at 0.5 (gates not learning)
- AMP scaler collapsing

9. Quick Reference Tables

9.1 Metric Health Summary

Metric	Desired Direction	Healthy Range
loss	↓	Config-dependent
mse	↓	
sharpe	↑	> 0 (ideally > 1)
npdd	↑	> 0
dd	↓	
turnover	↓	
fuzzy	Stable	0.001–0.01
rho	Stable	≈ 0.01
energy	↓	

9.2 Keyboard Shortcuts

Key	Action
Esc	Close modal
F12	Open browser dev tools

10. Troubleshooting Guide

10.1 Heatmap Not Updating

- Verify `/telemetry/fuzzy` endpoint activity.
- Ensure training script uses `--gui-telemetry`.
- Check browser console for WebSocket errors.

10.2 Metrics Frozen

- Training may be paused or crashed.
- Inspect `train.log`.
- Check GPU memory availability.

10.3 High Loss Values

- Learning rate may be excessive.
- Verify data normalization.
- Inspect debug output for NaN/Inf.

11. Conclusion

This report provides a structured, formal interpretation of the CondorBrain Training Dashboard. By understanding the behavior of predicate gates, loss components, diagnostics, and telemetry, operators can monitor training quality, detect instability, and ensure that CondorNet converges toward a stable, high-performance trading intelligence.